**Department of Physics and Astronomy**
**University of Heidelberg**

Bachelor Thesis in Physics
submitted by

**Anna V. Radoslavova**

born in Sofia (Bulgaria)

**2020**

# Using the FRG to Solve the Quantum-Mechanical Double-Well Potential and the O(N) Model in the Large N Limit in LPA

This Bachelor Thesis has been carried out by
**Anna V. Radoslavova**
at the Institute for Theoretical Physics, Heidelberg University
under the supervision of
**Prof. Dr. Jan M. Pawlowski**

# Abstract


**English:**

In the frame of this work a powerful tool for non-perturbative computations of the strong interactions is investigated. The functional renormalisation group FRG is applied to the anharmonic oscillator and the $O(N)$ model in the Large N limit. The flow equation for the anharmonic oscillator is derived to first order derivative expansion and the flow equation for the $O(N)$ model - to leading order derivative expansion. The derived flow equations are then solved in the local potential approximation LPA, using two different numerical approaches for the solution of the anharmonic oscillator. A local expansion and a PDE solutions are obtained. The energy gap is calculated for the PDE solution of the anharmonic oscillator, i.e. the energy difference between the ground state and the first excited state. A quantitative agreement for strong couplings is achieved, but for weak couplings - the solution shows large deviation. The solution of the $O(N)$ flow equation delivers qualitative results, showing the convexity restoring property of the flow.

**Deutsch:**

Im Rahmen dieser Arbeit beschäftigen wir uns mit einer nicht störungstheoretischen Methode für Berechnungen in dem Bereich der starken Kopplung. Die funktionale Renormierungsgruppe FRG wird auf den anharamonischen Oszillator und das $O(N)$ Model in dem Limit $N \to \infty$ angewandt. Die Flussgleichnung des anharmonischen Oszillators in der Ableitungsentwicklung erster Ordnung und die Flussgleichung des $O(N)$ Models in Ableitungsentwicklung nullter Ordnung werden anschließend hergeleitet. Die hergeleiteten Gleichnungen werden in LPA gelöst, indem für die Lösung der Flussgleichung des anharmonischen Oszillators zwei Methoden angewandt werden. Die zwei Lösungen, mittels der lokalen Entwicklung und der numerischen Lösung der partiellen Differentialgleichung PDG, werden erläutert. Für die Lösung der PDG wird die Energielücke, d.h. die Energiedifferenz zwischen dem Grundzustand und dem ersten angeregten Zustand, berechnet. Die Ergebnisse zeigen quantitative Übereinstimmung für den starkgekoppelten Bereich. Für schwachgekoppelte Systeme weisen die Ergebnisse jedoch eine große Abweichnung von den Sollwerten auf. Das Lösen der $O(N)$ Flussgleichung liefert qualitative Ergebnisse, die die Konvexitätwiederreproduzierung der Wetterichgleichung zeigen.

# Contents


**1. Introduction** **1**

**2. FRG and Statistical Physics Formulation** **3**

2.1. Motivation for the FRG . . . . . 3

2.2. Path Integral Formulation . . . . . 4

**3. Energy Gap and the Quantum Anharmonic Oscillator** **7**

**4. O(N) Model** **9**

**5. Derivation of the Flow Equation** **13**

**6. Derivation of the O(1) and O(N) Model Flow Equations** **17**

6.1. Flow Equation for Anharmonic Oscillator to First Order for $V(\phi)$ . . . . 17

6.2. Flow Equation for the $O(N)$ Model . . . . . 20

**7. Numerical Investigation** **23**

7.1. Polynomial Expansion . . . . . 23

7.2. Finite Difference Method . . . . . 25

**8. Results** **28**

8.1. Expansion for the Anharmonic Oscillator . . . . . 28

8.2. Solving the PDEs . . . . . 31

8.2.1. Anharmonic Oscillator . . . . . 31

8.2.2. Energy Gap . . . . . 33

8.2.3. Large N Limit $O(N)$ Model . . . . . 34

**9. Conclusion and Outlook** **36**

**A. Further Results** **38**

**B. 'Exact' Solution and Dilute Gas** **41**
B.1. Exact Numerical Solution . . . . . 41
B.2. Dilute Gas Instanton . . . . . 42

**C. Polynomial Expansion in $\phi$ to 12 Order** **43**

# Used Abbreviations and Notations

## Abbreviations

| | |
|---|---|
| **QFT** | Quantum Field Theory |
| **1PI** | One-Particle Irreducible |
| **LPA** | Local Potential Approximation |
| **FRG** | Functional Renormalisation Group |
| **ODE** | Ordinary Differential Equation |
| **PDE** | Partial Differential Equation |
| **QM** | Quantum Mechanics |
| **UV** | Ultraviolet |
| **IR** | Infrared |
| **QCD** | Quantum Chromodynamics |
| **RG time** | Renormalisation Group Time |
| **FDM** | Finite Difference Method |
| **RHS** | Right Hand Side |
| **VEV** | Vacuum Expectation Value |
| **PDG** | Partielle Differentialgleichung |
| **w.r.t.** | with respect to |

# Notations

The subscript k of the energy scale dependent quantities is omitted on some places for better overview and has to be understood implicitly. In the following some often used notations are given.

- For the local approximation of the $\phi$-dependent potential we use $a_{2n}$ for the running couplings with $n$ being a positive whole number and zero.
- For the rest of the thesis: $m_k^2$, $\lambda_k$ are used for the mass and interaction couplings.
- Compact notation for the derivatives for some arbitrarily chosen variables:

$$\begin{aligned} &\delta_\phi := \frac{\delta}{\delta\phi(x)}, \quad \partial_\mu := \frac{\partial}{\partial x_\mu}, \quad \partial_x := \frac{\partial}{\partial x}, \\ &V^{(2)}(\phi) := \frac{\delta^2 V(\phi)}{\delta\phi(x)\delta\phi(y)}, \quad V'(\rho) := \frac{\delta V(\rho)}{\delta\rho(x)}. \end{aligned} \tag{0.1}$$

- Derivatives of the potential:

$$V^{(1)} = \phi\, V', \quad V^{(2)} = V' + 2\rho\, V'', \quad \delta\phi = \phi\delta_\rho, \quad \text{with} \quad \rho := \frac{\phi^2}{2}. \tag{0.2}$$

- Abbreviations for the integrals:

$$\begin{aligned} &\int_p := \int \frac{d^d p}{(2\pi)^d} = \frac{\Omega_d}{(2\pi)^d}\int_0^\infty p^{d-1} dp \overset{d=1}{=} \frac{1}{\pi}\int_0^\infty dp\,, \quad \Omega_d = \frac{2\pi^{\frac{d}{2}}}{\Gamma\left(\frac{d}{2}\right)}, \\ &\int_x := \int d^d x = (2\pi)^d \delta(0) =: Vol_d \qquad\qquad \Omega_1 = 2. \end{aligned} \tag{0.3}$$

- Conventions for the Fourier transformations:

$$\phi(x) = \int_p \phi(p) e^{i\, p_\mu\, x_\mu}, \quad \phi(p) = \int_x \phi(x) e^{-i\, p_\mu x_\mu}. \tag{0.4}$$

# 1. Introduction

Our long-term goal is to find an approach that describes the quantum interactions using one generalised tool. A perturbative approach is not suitable for investigating theories based on large couplings. We want to study symmetry breaking, as this gives us more information on how a quantum system behaves. Therefore, the simplest toy model for investigating spontaneous symmetry breaking can be found in the anharmonic oscillator. Its potential is given by

$$V(\phi) = \frac{m^2}{2}\phi^2 + \frac{\lambda}{24}\phi^4. \tag{1.1}$$

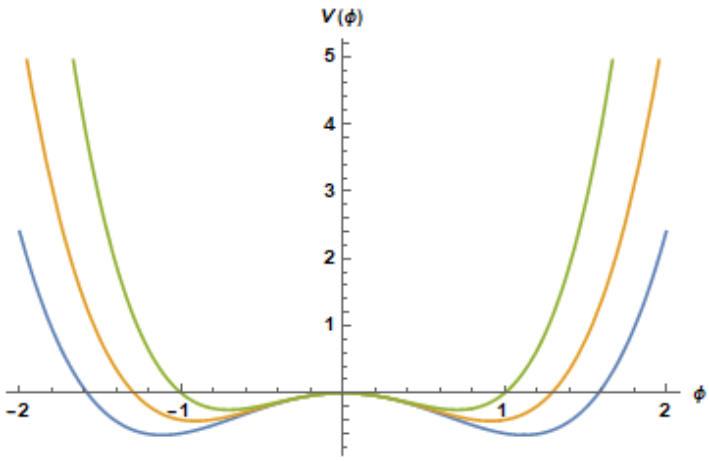


Figure 1.1.: Potential for Eq. 1.1 with $m^2 = -1$, antisymmetric case, for $\lambda_{norm} = 0.4$ (blue), $\lambda_{norm} = 0.7$ (yellow), $\lambda_{norm} = 1$ (green), with $\lambda_{norm} = \frac{\lambda}{24}$.

For positive values of the mass coupling $m^2$ we access the symmetric regime of the anharmonic potential which is well described by perturbation theory.
Therefore, the interesting case is obtained by going to negative values of $m^2$ as shown in Fig. 1.1. We get the double-well potential, which is prototypical for spontaneous symmetry breaking. It is important to understand this model, as it can be used for investigation of phase transitions [GW19], when the problem is translated into higher space-time dimensions.

Systems with continuous symmetries can experience symmetry breaking as well. For this reason, it is interesting for us to study such a system. To get more insights into the occurrence, we look at the $O(N)$ model[1]. If the anharmonic oscillator problem is treated classically, we would obtain two degenerate zero energy states. In quantum mechanics, however, there are also tunnelling solutions, obtained by the superposition of the two classical ground energy states. This gives rise to a characteristic difference between the ground state and the first excited state of the system. This difference is called *energy gap* and can be used for our purpose for investigating the accuracy of the functional renormalisation group, as the anharmonic oscillator has an analytic solution, coming from the stationary Schroedinger equation. More information can be found in Appendix B and [Son10].

In chapter 2 one can find a brief introduction to the FRG [Paw+18]. We proceed further with an actual computation of the $O(1)$ and Large N Limit of the $O(N)$ problems. The anharmonic oscillator can be described as $O(1)$ theory, where $O(1)$ states the symmetry group. In the case of only one scalar field the symmetry is discrete.

Furthermore, one can look at the large N limit of the $O(N)$ model [PS95], $N \to \infty$. The physical region is given rather by the $O(1)$ description.

We will see that the leading order derivative expansion that we use for our solution does not deliver good overlap with the 'exact' solution from the diagonalisation of the Hamiltonian for the weak coupling region. This is expected for the leading order result. We give the derivation of the potential flow equation to first order and set the wave function renormalisation constant to one and restore the equation for the leading order. The interested reader can find further information about different values for the wave function renormalisation in [Zap01][Wüs19].

We present two approaches for obtaining the effective potential - by a suitable expansion series and via solving the PDE numerically. In chapter 5 the chosen regulator is the Litim[2] regulator [Lit02], that is known to deliver simple and correct expressions for numerical investigations. At last, we compare the energy gap solutions for the FRG with the 'exact'[3] solution and look at their properties [Aok+98].

[1] The $O(N)$ model also called the linear sigma model.
[2] Also called flat or optimised regulator.
[3] The quotation marks mean exact numerical solution via diagonalisation of the Hamiltonian.

# 2. FRG and Statistical Physics Formulation

The FRG method for investigating strong coupled systems, which is known to work well is motivated in this chapter. Quantum chromodynamics QCD in the IR assumes large coupling values, which makes this regime inaccessible for perturbation theory. One method that is known to be well-working, is the functional renormalisation group, which is entirely non-perturbative. In the following a brief motivation from a statistical physics perspective is presented and related to the path integral formulation.

## 2.1. Motivation for the FRG

Quantum field theory is in a sense equivalent to statistical physics [Wip13]. This statement we make clear in this section. In statistical physics the object that gives us the important information about a system is the partition function. The basic idea is that the partition function is based on probabilities that tell us how likely it is to find a particle in a given energy state. The analog to that we find in the path integral formulation. It describes a sum over all possible paths that the fields can take in order to get to their final state. The two approaches are based on probabilities and they have their minor differences in the definition of variables. We will take the two formulations and show heuristically their equivalence. A more detailed derivation can be found in [PS95].
Assuming we have a given set of random variables $X_i$, having values $x_i$ and Hamiltonian $H(x_1, x_2, ...)$, then we can define our partition function as

$$Z(\beta) = \sum_{x_i} exp\big(-\beta H(x_1, x_2, ..)\big), \tag{2.1}$$

where $H$ is a real function on the space of states $X_1, X_2, ...$ and $\beta$ is a real free parameter, usually the inverse product of the temperature with the Boltzmann constant. The summation over $x_i$ is a sum over all possible values of each of the random variables $X_i$.

In continuous space the partition function can be written as an integral over these states for continuous random variables $X_i$[1]

$$Z(\beta) = \int_{x_i} exp\big(-\beta H(x_1, x_2, ..)\big). \tag{2.2}$$

If we want to have a more general expression and let $X_i$ be uncountable, we can work with functional integrals

$$Z = \int \mathcal{D}\varphi\, exp\big(-\beta H[\varphi]\big), \qquad \text{with} \qquad \mathcal{D}\varphi \equiv \prod_i d\varphi_i. \tag{2.3}$$

That formulation looks very similar to the one used in QFT as we shall see.

## 2.2. Path Integral Formulation

We start by introducing the energy scale k and the ultraviolet UV cutoff scale $\Lambda$. The k scale is used to take care of the infrared IR divergences, whereas $\Lambda$ removes the UV divergences. The partition function can be extended by adding auxiliary functions in the exponential function which do not change our physical dynamics. The functional integral formulation enables us to experiment with additional fields [2] which are the core of the path integral formulation.
An additional term, containing an external current $J$, is added, which is used to obtain the correlation functions by taking functional derivatives. There is a second term that is scale dependent and its role is to set an IR cutoff, below which the momentum modes are not integrated out[3]. In the following one can see the path integral formulation used in this thesis with the common notation[4]

$$\begin{aligned}\mathcal{Z}_k[J] &= \frac{1}{\mathcal{N}} \int [d\varphi]_{\text{ren}}\, exp\bigg(-S[\varphi] + \int_x J(x)\varphi(x) - \Delta S_k[\varphi]\bigg) \\ &= \frac{1}{\mathcal{N}} \int [d\varphi]_{\text{ren},\, p^2 \geq k^2}\, exp\bigg(-S[\varphi] + \int_x J(x)\varphi(x)\bigg).\end{aligned} \tag{2.4}$$

[1] $\int_{x_i} \equiv \int dx_1 dx_2 ...$
[2] Also called external source.
[3] That is the Wilsonian idea of momentum shell integration.
[4] $[d\varphi]_{p^2 \geq k^2} \equiv \prod_{p^2 \geq k^2} d\varphi_p$ meaning a measure for translation invariant fields.
$\int [d\varphi]_{\text{ren},p^2 \geq k^2} = \int [d\varphi]_{\text{ren}} exp(-\Delta S_k[\varphi])$.
$\mathcal{N}$ is a normalisation, which is an additive constant to the Schwinger functional $W_k$ and thus not relevant.

The scale dependent term is used to characterise the system evolution and to deliver results for the effective theory, starting from the classical one. This is the underlying property behind every FRG formulated problem.
In the $\varphi^4$ theory, the $S[\varphi]$ is the bare action[5]

$$S[\varphi] = \int_x \left[ \underbrace{\frac{1}{2}\partial_\mu\varphi(x)\partial_\mu\varphi(x)}_{\text{kinetic term}} + \underbrace{\frac{m^2}{2}\varphi(x)^2 + \frac{\lambda}{4!}\varphi(x)^4}_{\text{potential}} \right], \qquad \text{with} \qquad \int_x = \int d^d x. \tag{2.5}$$

The correlation functions can be obtained by taking functional derivatives of the path integral w.r.t. the external source $J$. Thus, we get an expression for the n-point correlation function

$$\frac{1}{\mathcal{Z}[J]} \frac{\delta^n \mathcal{Z}}{\delta J(x_1)...\delta J(x_n)} = \langle \varphi(x_1)...\varphi(x_n) \rangle_J. \tag{2.6}$$

A simpler expression giving us the connected correlation functions directly is defined by the Schwinger functional

$$W[J] = \ln \mathcal{Z}[J]. \tag{2.7}$$

The disconnected diagrams are not included in the Schwinger functional. What is obtained, are all connected diagrams which are important for our theory

$$\frac{\delta^n W[J]}{\delta J(x_1)...\delta J(x_n)} = \langle \varphi(x_1)...\varphi(x_n) \rangle_{\text{con}}. \tag{2.8}$$

The equivalence with Eq. 2.6 is then shown.
It is convenient to introduce a functional which generates the 1PI diagrams and this happens to be the so-called *effective action*. It is obtained by a Legendre transformation of the Schwinger functional $W[J]$. With $J \leftrightarrows \phi$ and the mean field notation $\phi(x) := \langle \varphi(x) \rangle$ the effective action takes the following form

$$\Gamma[\phi] = \sup_J \left( \int_x J(x)\phi(x) - W[J] \right) = \int_x J_{\text{sup}}(x) - W[J_{\text{sup}}]. \tag{2.9}$$

The Schwinger functional is a convex functional and use the current $J_{\text{sup}}$ in order to maximise the Legendre transform [JKM17] the subscript *sup* is dropped further in the thesis but kept in mind.
A graphical description of the effective action in theory space can be seen in Fig. 2.1, stating the fundamental idea of the computation.

[5] Also called classical action: $S_{bare} = \Gamma_{k\to\Lambda}$ action.

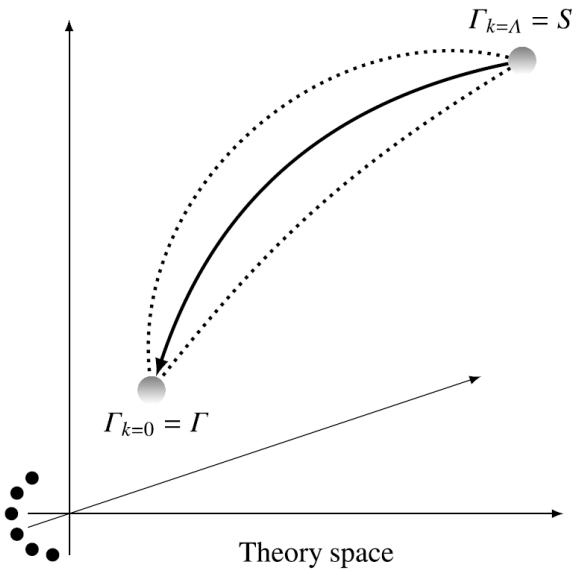


Figure 2.1.: Evolution of the action in dependence of the energy scale parameter k and the different operators of $\Gamma_k$, e.g. $(\partial\phi)^2$, $\phi^2$, $\phi^4$ [Wip13]. The start and end point of the effective action are fixed: $\Gamma_{k=\Lambda} = S_{bare}$, $\Gamma_{k=0} = \Gamma_{eff}$.

The bare and the effective actions do not depend on the scale parameter $k$. Nevertheless, the flow is scale dependent and can be adjusted in order to obtain simpler equations. When using different regulators, the action takes different paths approaching the effective action. The physics is in general not influenced by the usage of different regulators [Son10].

# 3. Energy Gap and the Quantum Anharmonic Oscillator

In this chapter a brief motivation for the occurrence of the energy gap[1] is given. We start with the potential of the anharmonic oscillator which is let to be implicitly scale dependent via the running couplings

$$V_k(\phi) = \frac{1}{2} m_k^2 \phi^2 + \frac{1}{24} \lambda_k \phi^4. \tag{3.1}$$

The action of the anharmonic oscillator to first order derivative expansion can be written as

$$\Gamma_k[\phi] = \int_x \left[ \frac{1}{2} Z_k(\phi) \partial_\mu \phi \partial_\mu \phi + V_k(\phi) + \mathcal{O}(\partial^4) \right]. \tag{3.2}$$

Note that Eq. 3.2 is given to first order derivative expansion in order to look at the dependence of the energy gap on the wave function renormalisation $Z_k$. For the actual computation in chapter 6 the wave function is set to one.
The mass coupling is obtained by taking two functional derivatives w.r.t. $\phi$

$$\begin{aligned}
\Gamma_k^{(2)}[\phi](p,q) &= \frac{\delta^2}{\delta\phi(p)\delta\phi(q)} \int_x \left[ \frac{1}{2} Z_k(\phi) \left( \int_{k_1} \partial_x \phi e^{ik_1 x} \right) \left( \int_{k_2} \partial_x \phi e^{ik_2 x} \right) + V_k(\phi) \right] \\
&= \frac{\delta^2}{\delta\phi(p)\delta\phi(q)} \int_x \left[ \frac{1}{2} Z_k(\phi) \left( \int_{k_1} ik_1 \phi(k_1) e^{ik_1 x} \right) \left( \int_{k_2} ik_2 \phi(k_2) e^{ik_2 x} \right) + V_k(\phi) \right] \\
&= -\int_x \left[ \int_{k_1} (2\pi)^d \delta(k_1 - q) \int_{k_2} (2\pi)^d \delta(k_2 - p) Z_k(\phi) k_1 k_2 + V_k^{(2)}(\phi) \right] e^{i(k_1+k_2)x} \\
&= \underbrace{\left( Z_k(\phi) p^2 + V_k^{(2)}(\phi) \right)}_{\Gamma_k^{(2)}[\phi](p)} (2\pi)^d \delta(p+q).
\end{aligned} \tag{3.3}$$

The computation is chosen to be in momentum space and therefore, a Fourier transformation of the fields is needed. In the second line in Eq. 3.3 the derivatives of the

[1] Also called mass gap.

exponential functions are performed and in the third line the functional derivatives are taken. The $\delta$-functions over the momenta and the additional normalisation factors of $(2\pi)^d$ come from being in momentum space. The functional derivatives of the potential are computed as well.
We arrive at the needed expression for $\Gamma_k^{(2)}[\phi](p)$

$$\Gamma_k^{(2)}[\phi](p) = Z_k(\phi)p^2 + V_k^{(2)}(\phi). \tag{3.4}$$

The second derivative of the effective action w.r.t. $\phi$ is the inverse propagator $G(x, y)$

$$G(x, y) = \frac{\delta^2 W[J]}{\delta J(x)\delta J(y)} = \frac{1}{\Gamma^{(2)}(x, y)}. \tag{3.5}$$

Having a pole mass, we get the needed expression for the energy gap

$$\begin{aligned} Z_k(\phi)p^2 + V_k^{(2)}(\phi) &\stackrel{!}{=} 0, \\ \Longrightarrow \Delta E_{gap} &= \sqrt{\frac{V_k^{(2)}(\phi)}{Z_k(\phi)}}\Bigg|_{\phi=\phi_0}. \end{aligned} \tag{3.6}$$

Wick rotation is needed in order to go back to Minkowski space, as in Euclidean space the time is imaginary. The energy gap is evaluated on the final convex potential where $\phi_0 = 0$ for the one-dimensional case which means no real symmetry breaking is present. The reason is that we have a vanishing vacuum expectation value for the ground state. In this case the energy gap plays a role of a renormalised mass!

# 4. O(N) Model

After the discussion of the anharmonic oscillator in chapter 2 we proceed with an investigation of the $O(N)$ symmetric models. They show a continuous symmetry and can be used to study spontaneous symmetry breaking. In the following a quick overview is presented, for more detailed information the reader is referred to [PS95] and [PJ20].

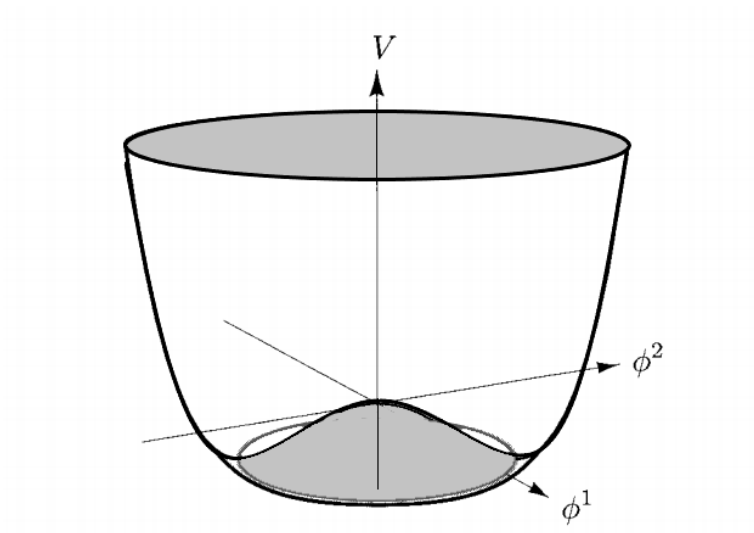


Figure 4.1.: Potential for spontaneous symmetry breaking for the special case of $N = 2$ of the $O(N)$ model.

Let us start with the quantum effective action of the $\phi^4$-theory. We want to take a look at an action, that has the same underlying structure as the anharmonic oscillator. For this purpose we take a $\vec{\phi}$-vector with its components being N-real scalar fields: $\phi_a$, $a \in [1, 2, \ldots, N]$. The k-dependent action takes the following form in leading order derivative expansion[1]

$$\Gamma_k[\phi] = \int_x \frac{1}{2}\partial_\mu\phi_a(x)\partial_\mu\phi_a(x) + V_k(\phi) + \mathcal{O}(\partial^2). \tag{4.1}$$

[1]Known as local potential approximation LPA.

It is important to point out that the quantum action is dependent on the mean field $\langle\varphi(x)\rangle = \phi(x)$. The effective potential is given by the extended version for multiple fields of the potential of the anharmonic oscillator. We assume a potential of the form

$$V_k(\phi) = \frac{1}{2}m_k^2\phi_a\phi_a + \frac{1}{4!}\lambda_k(\phi_a^2)^2. \tag{4.2}$$

The mass $m_k^2$ and the interaction couplings $\lambda_k$ are energy scale dependent. In Eq. 4.1 there is no single $\phi_a$ in the equation and the action only depends on the $\phi_a^2$ operators which means that the length for $\vec{\phi}$ is specified, but its direction is arbitrary.
For this reason the most convenient one is chosen for our further computation. Using the O(N)-symmetry of the system, we can rotate our $\phi$-vector

$$\phi_a(x) \to R_{ab}\phi_b(x), \qquad \text{with} \qquad R \in O(N). \tag{4.3}$$

The mass is again chosen to be negative in order to obtain a potential with a non-trivial minimum $\mu^2 := m_k^2$, with $\mu^2 < 0$. For the non-vanishing contribution of the $\vec{\phi}$ at the minimum the Nth-component is taken

$$\phi_{min} = (0, \ldots, 0, v), \qquad \text{with} \qquad v^2 = -\frac{6\,m_k^2}{\lambda_k} = -\frac{6\,\mu^2}{\lambda_k}. \tag{4.4}$$

The vacuum expectation value $v$ is calculated by taking a derivative of the potential and solving the equation at the minimum. Because of the $O(N)$-symmetry we have

$$\left.\frac{\partial V_k(\phi)}{\partial \phi_a}\right|_{\phi^R_{min}=R\phi_{min}} = 0. \tag{4.5}$$

We proceed with a further expansion around that chosen minimum where the common notation for association of the scalar fields is used as follows[2]

$$\phi(x) = (\pi_1, \ldots, \pi_{N-1}, v + \sigma(x)). \tag{4.6}$$

With Eq. 4.1 we get after algebraic manipulation an expression which does not contain

[2]More on the notation in section 6.2

a mass term for the $\pi^k$ scalar fields $k \in [1, \dots N-1]$

$$\Gamma_k[\vec{\pi}, \sigma] = \int_x \left[ \frac{1}{2}(\partial\sigma)^2 + \frac{1}{2}(2\,\mu^2)\sigma^2 + \frac{1}{2}(\partial\vec{\pi})^2 + v\sigma\vec{\pi}^2 + v\sigma^3 + \left( \frac{\lambda}{4!}(\vec{\pi}^2)^2 + \sigma^4 + 2\vec{\pi}^2\sigma^2 \right) \right]. \tag{4.7}$$

The symmetry of the problem has changed to a $O(N-1)$ symmetry due to the mass term of the $\sigma$-field. Counting the generators of the groups, the number # of the broken symmetries is found to be $N-1$

$$\begin{aligned} \text{\# of generators for O(N)} &\equiv \frac{N\,(N-1)}{2}, \qquad (4.8) \\ \text{\# of generators for O(N-1)} &\equiv \frac{(N-1)\,(N-2)}{2}, \\ \implies \text{\# of generators for O(N)} - \text{\# of generators for O}(N-1) &= \underbrace{N-1.}_{\text{\# of Goldstone bosons}} \end{aligned}$$

With the assumption for constant fields[3], the effective action can be split in parts dependent on the $\phi$-derivatives and rest. The terms proportional to the derivatives of $\phi$ are set to zero

$$\Gamma_k[\phi] = \underbrace{\Gamma_0[\partial\phi]}_{=0} + \underbrace{\int_x}_{Vol_d} V_k(\phi(x)). \tag{4.9}$$

We are dealing with continuous transformations and can then rotate $\phi$

$$\phi \to \phi + \epsilon\alpha(\phi). \tag{4.10}$$

In the $O(N)$ we have

$$\alpha_a(\phi) = \omega_{ab}\phi_b. \tag{4.11}$$

The symmetry requires the potential to be invariant under the infinitesimal transformation from Eq. 4.10

$$V_k(\phi + \epsilon\alpha(\phi)) \stackrel{!}{=} V_k(\phi). \tag{4.12}$$

Therefore, we expand the potential around the its minimum and get

$$V_k(\phi) = V_k(\phi_{min}) + \left.\frac{\partial V_k(\phi)}{\partial\phi_b}\right|_{\phi_{min}} (\phi_b - \phi_{min}) + \frac{1}{2}\left.\frac{\partial V_k(\phi)}{\partial\phi_a\partial\phi_b}\right|_{\phi_{min}} (\phi_a - \phi_{min})(\phi_b - \phi_{min}) + \mathcal{O}(\phi^3). \tag{4.13}$$

[3]Constant fields means, assuming translation and Lorentz invariant objects.

The first derivative vanishes because of the choice of our expansion point. Consequently, the term with the second derivative needs to vanish too for the symmetry to be preserved

$$\alpha_a\Big|_{\phi_{min}} \frac{\partial^2 V_k(\phi)}{\partial\phi_a\partial_b\phi}\Bigg|_{\phi_{min}} = 0. \tag{4.14}$$

For $\alpha_a(\phi_{min}) \neq 0$:

$$\implies \frac{\partial^2 V_k(\phi)}{\partial\phi_a\partial_b\phi}\Bigg|_{\phi_{min}} \stackrel{!}{=} 0.$$

Going back to Eq. 4.2, the second $\phi$-derivative at the minimum gives the mass of the corresponding particle which needs to vanish here. Thus, we have massless particles, called *Goldstone bosons*. The essence of the Goldstone theorem can be seen in equations Eq. 4.9 to Eq. 4.14, stating that for every broken symmetry a massless particle, a Goldstone boson, is present.

# 5. Derivation of the Flow Equation

In this chapter an instructive derivation of the Wetterich equation can be found which is used for our further investigation of the anharmonic oscillator and the $O(N)$ model. The path integral is let to be dependent on the energy scale (see Eq. 2.4).
The k-dependent Schwinger functional $W_k[J]$ takes the form

$$W_k[J] = \ln \mathcal{Z}_k[J]. \tag{5.1}$$

The additional regulator dependent function can be chosen to be either in position or in momentum space. Starting from position space, the equation yields

$$\Delta S_k = \frac{1}{2} \int_x \int_y \phi(x)\, R_k(x,y)\, \phi(y). \tag{5.2}$$

Useful relations for the regulator are given in position and momentum space

$$R_k(x,y) = R_k(-\partial_x^2)\, \delta(x-y). \tag{5.3}$$

Applying Fourier transformation, we get

$$R_k(p,q) = R_k(p)(2\pi)^d \delta(q+p). \tag{5.4}$$

The Legendre transformation of the Schwinger functional renders

$$\Gamma_k[\phi] = \sup_J \left( \int J\phi - W_k[J] \right) - \Delta S_k[\phi]. \tag{5.5}$$

The $\Delta S_k[\phi]$ is implicitly energy scale dependent, via the regulator function $R_k$.
The connected 2-point function[1] is also needed for our purpose. In position space using

[1] Also the propagator for $J(x)$ and $J(y)$.

Eq. 2.8 it yields

$$G_k(x,y) = \frac{\delta^2 W_k[J]}{\delta J(x)\delta J(y)} = \langle\varphi(x)\varphi(y)\rangle - \langle\varphi(x)\rangle\langle\varphi(y)\rangle. \tag{5.6}$$

For the propagator in momentum space after Fourier transformation the equation renders

$$G(p,q) = G(p,-p)(2\pi)^d\delta(p+q). \tag{5.7}$$

Finally, the modified partial derivative w.r.t. the energy scale k, $k\partial_k = \partial_t$ with $t$ being the renormalisation group RG time, can be applied[2]. The expression for the RG time derivative of the Schwinger functional contains only the k-dependent contribution

$$\begin{aligned}\partial_t W_k[J] = \partial_t \ln\mathcal{Z}_k[J] &= \frac{1}{\mathcal{Z}_k[J]}\partial_t\mathcal{Z}_k[J] \\ &= -\frac{1}{2}\int_x\int_y \langle\varphi(x)\partial_t R_k(x,y)\varphi(y)\rangle \\ &\overset{Eq.\ 5.6}{=} -\frac{1}{2}\int_x\int_y \partial_t R_k(x,y)G_k(x,y) - \partial_t\Delta S_k[\phi].\end{aligned} \tag{5.8}$$

The RG time derivative is also applied on the modified Legendre transformation (see Eq. 5.5), we obtain

$$\partial_t\Gamma_k[\phi] = \int_x \partial_t J(x)\phi(x) - \partial_t W_k[J] - \int_x \underbrace{\frac{\partial W_k[J]}{\partial J(x)}}_{\phi(x)}\partial_t J(x) - \partial_t S_k[\phi]. \tag{5.9}$$

Using

$$\phi(x) = \frac{\partial W_k[J]}{\partial J(x)}. \tag{5.10}$$

The RG time derivative of the 1PI generator function takes the following form

$$\partial_t\Gamma_k[\phi] = -\partial_t W_k[J] - \partial_t S_k[\phi]. \tag{5.11}$$

Comparing Eq. 5.8 with Eq. 5.11, the expression for the RG time derivative of the 1PI generator yields

$$\partial_t\Gamma_k[\phi] = \frac{1}{2}\int_x\int_y \partial_t R_k(x,y)G_k(x,y). \tag{5.12}$$

[2]Note, a slightly different convention is used in chapter 7.

The Wetterich equation in its commonly given form renders

$$\partial_t \Gamma_k[\phi] = \frac{1}{2}\mathrm{tr}\, \partial_t R_k G_k = \frac{1}{2}\mathrm{tr}\, \frac{\partial_t R_k}{\Gamma_k^{(2)}[\phi] + R_k}. \tag{5.13}$$

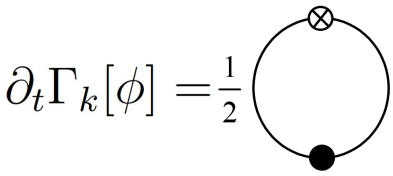


Figure 5.1.: Diagrammatic representation of the Wetterich equation. The crossed circle represents the RG time derivative of the regulator and the filled dot stands for the propagator.

In Eq. 5.13 the propagator relation Eq. 5.14 is used. The position arguments are left out for more compact notation

$$G_k = \frac{1}{\Gamma_k^{(2)}[\phi] + R_k}. \tag{5.14}$$

Eq. 5.8 in its energy scale dependent form is the Polchinski equation given by

$$\partial_k W_k[J] = \frac{1}{2}\int_x \int_y \partial_k R(x,y) G_k(y,x) - \partial_k \Delta S_k[\phi]. \tag{5.15}$$

The final expression for the Wetterich equation is an exact differential equation despite the 1-loop structure. That is an impressive result, but due to the full propagator, entering the PDE, not a surprising one.
Until this point of the investigation our regulator is in its general form. Now it can be chosen freely, as long as it obeys the following conditions

$$\begin{aligned} &\lim_{p^2/k^2 \to 0} R_k(p) > 0, \\ &\lim_{k^2/p^2 \to 0} R_k(p) = 0, \\ &\lim_{k \to \Lambda \to \infty} R_k(p) = \infty. \end{aligned} \tag{5.16}$$

The given conditions in Eq. 5.16 take care of the correct limits of the RG time dependent functional. They restore the bare action for large momenta and reach the effective action for the limit $k \to 0$. The conditions are given in momentum space, as this is the typical

notation, given also in the [Paw+18][Aok+98].
Given the conditions, the Litim regulator is a common choice, also used here. It obeys the stated conditions and provides us with the necessary stability while pursuing numerical investigation

$$R_k(p) = Z_k(k^2 - p^2)\Theta(k^2 - p^2), \tag{5.17}$$

where $Z_k$ is a function which can be chosen for our purpose. In leading order derivative expansion of the scale dependent action, $Z_k \equiv 1$, as we do not have a dependence of the effective action on the wave function renormalisation.

# 6. Derivation of the O(1) and O(N) Model Flow Equations

In the following, the simplified expressions solved in chapter 8 are derived. We first proceed with the derivation of the potential flow equation for the anharmonic oscillator to first order derivative expansion as we did in chapter 3. One can get more insights into how the energy gap for the weak couplings gets its wave function renormalisation dependence. At last, we derive the flow equation for the O(N) model and take its large N limit.

## 6.1. Flow Equation for Anharmonic Oscillator to First Order for $V(\phi)$

Let us start with the effective action to first order derivative expansion. It is given again explicitly, as this is one of the most important assumptions for the solution

$$\Gamma_k[\phi] = \int_x \left[\frac{1}{2} Z_k(\phi)\partial_\mu\phi\partial_\mu\phi + V_k(\rho) + \mathcal{O}(\partial^4)\right]. \tag{6.1}$$

Using the Wetterich equation from Eq. 5.13 with the assumption of constant fields, it can be written in terms of the potential as follows

$$\partial_t\Gamma[\phi]|_{\phi=const} = \int_x \partial_t V(\phi) = \partial_t V(\phi) Vol_d. \tag{6.2}$$

We still need the RG time derivative of the regulator, given in Eq. 5.17, for our further computations

$$\partial_t R_k(p) = \partial_t Z(\phi)(k^2 - p^2)\Theta(k^2 - p^2) + 2Z(\phi)k^2\Theta(k^2 - p^2). \tag{6.3}$$

The derivative of the Heaviside function proportional to $\delta(k^2 - p^2)$ is left out due to the symmetry of the problem, as the term is simply zero.
Using the derived Wetterich equation in Eq. 5.12, we can get rid of the volume factors in Eq. 6.2 by using Eq. 3.3. The approximation, however, delivers good results in our setup, but it is in general not entirely correct[1].
One can write the simplified flow equation for the scale dependent potential using Eq. 5.14, Eq. 5.4 and Eq. 5.7 for the regulator and the propagator

$$\begin{aligned}\partial_t V_k(\phi) &= \frac{1}{2}\int_{-k}^{k} \frac{d^d p}{(2\pi)^d} \frac{\partial_t Z(\phi)(k^2 - p^2) + 2Z(\phi)k^2}{Z(\phi)p^2 + V^{(2)}(\phi) + Z(\phi)\left(k^2 - p^2\right)} \\ &\overset{\mathrm{d}=1}{=} \frac{1}{3\pi} \frac{(\partial_t Z(\phi) + 3Z(\phi))k^3}{Z(\phi)k^2 + V^{(2)}(\phi)}. \end{aligned} \tag{6.4}$$

The first line shows the multidimensional case of the flow equation where we can easily obtain the equation for any needed dimension. We now see that for setting $Z \equiv 1$, we get our simplified flow equation for the anharmonic oscillator to leading order in the derivative expansion

$$\partial_t V_k(\phi) = \frac{1}{\pi} \frac{k^3}{k^2 + V^{(2)}(\phi)}. \tag{6.5}$$

So far the minimum of the potential in the flow equation is not specified. This is something which is done to simplify or otherwise give a more natural evolution of the equations. It is convenient to have a vanishing flow for the minimum of the potential

$$\begin{aligned}\partial_t V_k(\phi) &= \frac{1}{\pi} \frac{k^3}{k^2 + V^{(2)}(\phi)} - \frac{1}{\pi} \frac{k^3}{k^2 + V^{(2)}(\phi_{min})} \\ &= \frac{1}{\pi} \frac{k^3}{k^2 + V'(\rho) + 2\rho V''(\rho)} - \frac{1}{\pi} \frac{k^3}{k^2 + V'(\rho_{min}) + 2\rho_{min} V''(\rho_{min})}. \end{aligned} \tag{6.6}$$

For one dimension[2] we have $\rho_{min}$ for the effective potential at zero[3]. This is due to the fact that there is only one dimension. With all that being said, our final result for the

[1] We can do calculation for changing fields, but this is of course far more complicated mathematically and numerically. This won't be dicussed further in this thesis.
[2] This is nothing, but simply the quantum mechanical case.
[3] For the $\phi$-dependent case the same argument holds.

flow equation of the scale dependent potential reads

$$\begin{aligned}\partial_t V_k(\phi) &= \frac{1}{\pi}\frac{k^3}{k^2+V^{(2)}(\phi)} - \frac{1}{\pi}\frac{k^3}{k^2+V^{(2)}(\phi_{min})} \\ &= \frac{k}{\pi}\left(\frac{k^2}{k^2+V'(\rho)+2\rho V''(\rho)} - 1\right). \end{aligned} \tag{6.7}$$

We proceed with the numerical investigation in chapter 8.

## 6.2. Flow Equation for the $O(N)$ Model

The derivation of the flow for the $O(N)$ model is conceptually analogous to the derivation of the flow for the anharmonic oscillator. In contrast to the $O(1)$ model, the symmetry is continuous and there are more fields that respect the symmetry. In the following the notation given in chapter 4 is used. For the flow equation of the scale dependent potential we need the second functional derivative w.r.t. the fields and the RG time derivative of the regulator, given in Eq. 6.3. Then we proceed analogously to section 6.1.
The second functional derivative of the effective action for $O(N)$ models is calculated in the following

$$\begin{aligned}
\Gamma^{(2)}[\phi](p,q)_{b,c} &= \frac{\delta^2\Gamma}{\phi_b(p)\phi_c(q)} = \frac{\delta^2}{\phi_b(p)\phi_c(q)}\int_x\left(\frac{1}{2}(\partial_x\phi_a)^2+V\right)\\
&= \frac{\delta^2}{\phi_b(p)\phi_c(q)}\int_x\left(\frac{1}{2}\partial_x\phi_a(x)\partial_x\phi_a(x)+V\right)\\
&= \frac{\delta^2}{\phi_b(p)\phi_c(q)}\int_x\Big(\frac{1}{2}\Big(\int_{k_1}\partial_x\phi_a(k_1)e^{ik_1x}\Big)\Big(\int_{k_2}\partial_x\phi_a(k_2)e^{ik_2x}\Big)+V\Big)\\
&= \frac{\delta^2}{\phi_b(p)\phi_c(q)}\int_x\Big(\int_{k_1}\int_{k_2}-\frac{1}{2}k_1\phi_a(k_1)k_2\phi_a(k_2)e^{i(k_1+k_2)x}+V\Big)\\
&= \int_x\Bigg(\int_{k_1}\int_{k_2}-k_1k_2(2\pi)^d\delta(k_1-p)(2\pi)^d\delta(k_2-q)e^{i(k_1+k_2)x}\delta_{ab}\delta_{ac}+\\
&+\Big(V'\delta_{bc}+\phi_b\phi_cV''(\rho)\Big)e^{i(p+q)x}\Bigg)\\
&= \int_x\Big(-pq+V')\delta_{bc}+\phi_b\phi_cV''(\rho)\Big)e^{i(p+q)x}\\
&= \Big(-pq+V'\delta_{bc}+\phi_b\phi_cV''(\rho)\Big)(2\pi)^d\delta(p+q)\\
&= \Bigg(\Big(p^2+V'(\rho)\Big)\delta_{bc}+\phi_b\phi_cV''(\rho)\Bigg)(2\pi)^d\delta(p+q).
\end{aligned}\tag{6.8}$$

We use Fourier transformation which enables us to work with simpler expressions. The computation is straightforward, we get the $\delta_{ab}$, $\delta_{ac}$ functions assuring, that we operate on the same scalar fields. The momentum $\delta$-function take care of the energy conservation. Analogously to Eq. 3.3 we get

$$\Gamma^{(2)}[\phi](p)_{b,c} = \left(\left(\frac{p^2+q^2}{2}+V'(\rho)\right)\delta_{bc}+\phi_b\phi_cV''(\rho)\right). \tag{6.9}$$

Going back to the Wetterich equation, we can calculate the flow equation

$$\begin{aligned}\partial_t V_k(\rho) &= \frac{1}{2}\int_p G_k(p)\,\partial_t R_k(p)\\ &= \frac{1}{2}\int_p \frac{1}{\Gamma^{(2)}[\phi](p)+R_k}\partial_t R_k\\ &= \frac{1}{2}\int_p \frac{1}{\big(p^2+V'(\rho)\big)\delta_{bc}+\phi_b\phi_c V''(\rho)+R_k}\partial_t R_k(p).\end{aligned} \tag{6.10}$$

The $\phi$-field can be oriented in a suitable for the computation direction. Thus, the Nth component is chosen to be the non-vanishing contribution. In general, every direction can be chosen, but as we shall see in the following, this constellation delivers conceptually easily understandable results.
We can write then $\phi_a = (0,..,\sigma)$, where we use the standard notation. The $\sigma$-direction is associated with the $\sigma$-meson and the $N-1$ components of $\vec{\phi}$ - with the 'pions'[4]. For $\Gamma^{(2)}[\phi](p)$ we obtain

$$\Gamma^{(2)}[\phi](p) = \Big(p^2+V'(\rho)\Big)\delta_{ab}+\phi_a\phi_b V''(\rho)\delta_{aN}\delta_{bN}. \tag{6.11}$$

From Eq. 6.11 the $\phi_a\phi_b$-term gets its contribution only for the Nth component because of our choice of orientation: This is exactly the mass direction of the $\sigma$-meson. For the Wetterich equation, after performing the summation over the components, it follows

$$\partial_t V_k(\rho) = \frac{1}{2}\int_p \left[\frac{N-1}{p^2+V_k'(\rho)+R_k(p)}+\frac{1}{p^2+V'(\rho)+2\rho V_k''+R_k(p)}\right]\partial_t R_k(p). \tag{6.12}$$

In order to obtain an equation which can be solved easier numerically or even analytically the substitution[5]: $V_k(\rho) = U_k(N-1)$ and $\rho = \rho'(N-1)$, is used

$$\begin{aligned}\partial_t U_k(N-1) = \frac{1}{2}\int_p (N-1)\partial_t R_k(p)\Bigg[&\frac{1}{p^2+\frac{\partial}{\partial\rho'(N-1)}U_k(N-1)-R_k(p)}\\ &+\frac{1}{N-1}\frac{1}{p^2+\frac{\partial}{\partial\rho'(N-1)}U_k(N-1)+2\rho'(N-1)\frac{\partial^2}{\partial\rho'^2(N-1)^2}U_k(N-1)}\Bigg].\end{aligned} \tag{6.13}$$

[4]In the special case of $O(4)$-theory we get the pion model with the three massless pions (Goldstone bosons).
[5]Achieving a simple rescaling.

After taking care of the rescaling factors $N-1$ and setting the relabeling of the new potential and fields back to $V_k$ and $\rho$ respectively we result in the following expression

$$\partial_t V_k(\rho) = \frac{1}{2}\int_p \left[\frac{1}{p^2 + V_k'(\rho) + R_k(p)} + \frac{1}{N-1}\frac{1}{p^2 + V'(\rho) + 2\rho V_k'' + R_k(p)}\right]\partial_t R_k(p). \tag{6.14}$$

With Eq. 6.3 for $Z \equiv 1$ and the limit $N \to \infty$ we get the simplified expression

$$\partial_t V_k = \frac{1}{2}\int_p \frac{2k^2\Theta(k^2-p^2)}{p^2 + V'(\rho) + (k^2-p^2)\Theta(k^2-p^2)}. \tag{6.15}$$

The Heaviside function assures that an IR cutoff is present. The momentum integration can be performed straightforwardly in our case, as there is no implicit dependence of the objects entering the equation. Our final expression for the large N-limit $O(N)$ contribution yields

$$\begin{aligned}\partial_t V_k(\rho) &= \int_{-k}^{k}\frac{dp}{2\pi}\frac{k^2}{p^2 + V'(\rho) + k^2 - p^2} \\ &= \frac{1}{\pi}\frac{k^3}{k^2 + V'(\rho)}.\end{aligned} \tag{6.16}$$

Here the minimum of the flow equation can be chosen and the assumptions for the $O(1)$ case hold[6]. Additionally, due to the Mermin-Wagner-theorem a symmetry breaking is not possible for systems having continuous symmetry in $d = 1$ and $d = 2$. Consequently, our minimum is found at $\rho_{min} = \phi_{min} = 0$.

[6]For more information see Eq. 6.6 and argumentation

# 7. Numerical Investigation

In the following an explanation on how the derived in chapter 6 equations for the effective potential are solved can be found. In section 7.1 an insight is given, into how we solve the equations for the anharmonic oscillator, using an expansion ansatz for Crude truncation[1]. The next few orders are given in Appendix C. We then proceed by introducing a more advanced method for solving the flow equation - the Finite Difference Method - and use Python to implement it.

## 7.1. Polynomial Expansion

A well-working simple technique - an expansion scheme - can be used in order to simplify the calculation since our potential takes such a suitable form.
For our aim we can use a polynomial expansion

$$V(\phi) = \sum_{n=0}^{\infty} \frac{a_{2n,k}\,(\phi - \phi_{min})^{2n}}{(2n)!} \equiv V(\rho) = \sum_{n=0}^{\infty} \frac{\lambda_{n,k}\,(\rho - \rho_{min})^{n}}{n!}. \tag{7.1}$$

The coefficients $a_{2n,k}$ and $\lambda_{n,k}$ play the role of the running couplings - for the $\phi$ and $\rho$-convention respectively. We then truncate the system at $n = 2$ and thus, we get a polynomial function of the form

$$V(\phi) = a_0 + \frac{1}{4}a_2\phi^2 + \frac{1}{24}a_4\phi^4 = \lambda_0 + \lambda_1\rho + \frac{1}{2}\lambda_2\rho. \tag{7.2}$$

This enables us to find the flow equation of the truncated system by taking functional derivatives of the RHS of the flowing equation Eq. 6.5 for the potential w.r.t $\phi$ or $\rho$ respectively depending on the used convention[2]. Doing so, we need to look at the relevant contributions only. We choose to perform the derivatives using Mathematica and set the $\phi's$ at the end to zero. That gives the right expressions. Nevertheless, we

[1]The couplings starting from $a_6$ are set to zero.
[2]In the following the $\phi$-dependent potential is used.

have to be careful when such a local approximation is used. The reason for this is that we have chosen only one expansion point, but we are seeking for more non-trivial minima at the end. This is, however, as already mentioned in chapter 6, not possible in our setup, but it is also not possible in general. Going to $\rho$-dependent potential, the system does not show such a limitation[3].

The running couplings' equations are calculated in dimensionful form for the $\phi$-choice of variables. Because of that, an expansion around $\phi_{min} = 0$ is taken. It shows rather good convergence properties for large starting values of the $a_4$-couplings for the anharmonic oscillator as we shall see.

The equations for the first 12 couplings can be found in Appendix C, but we calculate the coupled system of the first three first order ordinary differential equations ODE, using the scipy package odeint.

For the Crude truncation, $a_6 = ... = a_\infty = 0$, the first three coupled ODEs yield

$$\begin{aligned}
\partial_k a_0 &= -\frac{1}{\pi}\frac{a_2}{k^2 + a_2}, \\
\partial_k a_2 &= -\frac{1}{\pi}\frac{k^2\, a_4}{(k^2 + a_2)^2}, \\
\partial_k a_4 &= \frac{1}{\pi}\frac{6\, k^2\, a_4^2}{(k^2 + a_2)^3}.
\end{aligned} \tag{7.3}$$

The evolution of the couplings is once given in dependence on the scale parameter and then on the RG defined time. Here, the modified RG time is used which is a dimensionless quantity. The slight difference of the RG time conventions is denoted in chapter 6. The following modification is used to make the time evolution more natural and easier for numerical implementation[4]

$$\begin{aligned}
t_{\mathrm{RG}} &= -\ln\left(\frac{k}{\Lambda}\right), \\
\Longrightarrow\ k &= \Lambda e^{-t_{\mathrm{RG}}}.
\end{aligned} \tag{7.4}$$

[3]One can choose a suitable expansion point, that deliver even better results.
[4]The deviation is the minus sign in front of the logarithm.

The RG subscript is omitted in the following. The energy scale parameter $k$ is substituted by $t$ in our three coupled ODEs and the evolution of the couplings for the RG time reads

$$\begin{aligned}
\partial_t a_0 &= -\frac{1}{\pi}\frac{(\Lambda e^{-t})\, a_2}{(\Lambda e^{-t})^2 + a_2}, \\
\partial_t a_2 &= -\frac{1}{\pi}\frac{(\Lambda e^{-t})^3\, a_4}{((\Lambda e^{-t})^2 + a_2)^2}, \\
\partial_t a_4 &= \frac{1}{\pi}\frac{6\,(\Lambda e^{-t})^3\, a_4^2}{((\Lambda e^{-t})^2 + a_2)^3},
\end{aligned} \tag{7.5}$$

with the UV cutoff being $\Lambda = 1 \times 10^4$ the scale parameter $k$ starting at $k_{in} = \Lambda$ and flowing towards $k = 0$. For the implementation we need to consider the end point when using the RG time and let $k \to 0$, but not reaching the exact value, as we have not a defined RG time by that value of k. We set k to the minimal possible value which happens to be $k_{final} = 1 \times 10^{-16}$ due to numerical limitations.
The initial conditions for the couplings are to be chosen, as they define our bare action. Because we want to have non-trivial minima at the beginning, we choose for the initial mass coupling $a_2$ a negative value

$$\begin{aligned}
a_0 &= 0, \\
a_2 &= -1, \\
a_4 &= [0, 1, \ldots, 10].
\end{aligned} \tag{7.6}$$

Thus, in chapter 8 we get the flowing diagrams for the first three couplings[5]. The values for the couplings at $k = 0$ are read off, as there the final effective action is obtained.

## 7.2. Finite Difference Method

Here, a brief insight into one often used numerical technique for solving PDEs - the Finite Difference Method - is presented. Let us assume we have a continuous domain. For a numerical implementation, however, a discrete one is needed. For this reason, a set of equidistant points is introduced. The discretisation is used to approximate the second and first functional derivatives of the potential w.r.t. the field that come in the denominators of Eq. 6.5, Eq. 6.7 and Eq. 6.16.
FDM is presented in the one dimensional case, but it can be straightforwardly extended

[5]Note, that we present the solution for $a_4 = 3$ in chapter 8

to more dimensions. Let us assume we have a bounded domain $\Omega = [-L, L]$. The discretisation parameter is chosen to be $\phi$. We set the number of points to $\#\phi = N$. It can be used to define the step value $\Delta\phi$, as the evolution is convenient to get increased by the same value of the $\phi$-parameter each time

$$\begin{aligned}\phi_i &= -L + i\frac{L-(-L)}{N-1}, \qquad \text{with} \qquad i = 0, \dots N-1 \\ &= -L + i\frac{2L}{N-1} = -L + i\Delta\phi. \end{aligned} \tag{7.7}$$

The expression for the derivatives of a sufficiently differentiable function $v(t, \phi)$ dependent on $\phi$ and some other free variable t can be written as a discrete object

$$\begin{aligned}\text{First central derivative: } \partial_\phi v(t,\phi_i) &\approx \frac{v(t,\phi_{i+1}) - v(t,\phi_{i-1})}{2\Delta\phi}, \\ \text{First right-handed derivative: } \partial_\phi v(t,\phi_i) &\approx \frac{v(t,\phi_{i+1}) - v(t,\phi_i)}{\Delta\phi}, \\ \text{Second central derivative: } \partial_\phi^2 v(t,\phi_i) &\approx \frac{v(t,\phi_{i+1}) - 2v(t,\phi_i) + v(t,\phi_{i-1})}{\Delta\phi^2}. \end{aligned} \tag{7.8}$$

The FDM discretisation requires setting the function on the boundaries to zero, meaning $v(\phi_0) = v(\phi_{N-1}) = 0$[6].
A compact notation for the discretised functions of the $\phi$'s reads

$$V = (v_1(t), \dots v_{N-2}(t))^T. \tag{7.9}$$

For further numerical investigation the expressions of the derivatives can be written in a matrix [Pav19] form with M being a tridiagonal matrix[7]

$$\begin{aligned}\partial_\phi^2 V(\phi) &= -M(v_1(t), \dots v_{N-2}(t))^T + f(t,\phi) \\ &= \frac{1}{\Delta\phi^2}\begin{pmatrix} 2 & -1 & 0 & \dots & 0 \\ -1 & 2 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & -1 \\ 0 & \dots & \dots & -1 & 2 \end{pmatrix}\begin{pmatrix} v_1(t) \\ \vdots \\ \vdots \\ v_{N-2}(t) \end{pmatrix} + f(t,\phi), \\ \text{with} \qquad & f(t,\phi) = (f_1(t,\phi_1), \dots f_{N-2}(t,\phi_{N-2})^T. \end{aligned} \tag{7.10}$$

[6] Assures, that the starting point is sufficiently far away.
[7] The function $f(t, \phi)$ is not needed for our purpose and is part of the more general expression.

Using this method to solve our flow equations, a special care needs to be taken when the flow approaches the minimum of the potential in the $\rho$-substitution. There the boundaries for $k = 0$ cannot be specified, because we are seeking exactly this result for the effective potential. However, the problem can be avoided by using a right-handed derivative for the first $\rho$-derivative of the potential in the denominator and central derivative for the second $\rho$ derivative in Eq. 6.7. That stabilises the flow and a qualitative solution for the both $O(1)$ and Large N limit $O(N)$ model in one dimension can be obtained. More details can be found in chapter 8.

# 8. Results

In this chapter the solved graphical solution of the equations, discussed in chapter 6 and chapter 7, are presented. All of the diagrams are adapted for better visualisation[1].

## 8.1. Expansion for the Anharmonic Oscillator

An integration of the three ODEs for different initial values for the $\phi^4$ coupling $a_4$ is performed. The initial conditions for $a_0$ and $a_2$ remain unchanged throughout the variation of $a_4$. In this section the integrated equations for initial value $a_4 = 3$ are shown, since the equations get unstable for smaller values[2] and the obtained results are not reliable. However, for larger initial values we get plausible results. The symmetric case is also computed in order to see, whether there is a difference in the capabilities of this approach for different setups. Indeed, there is no critical value, beyond which the flow collapses. This is not surprising, as we get the harmonic potential for $a_4 = 0$. The graphical results are presented in Appendix A.

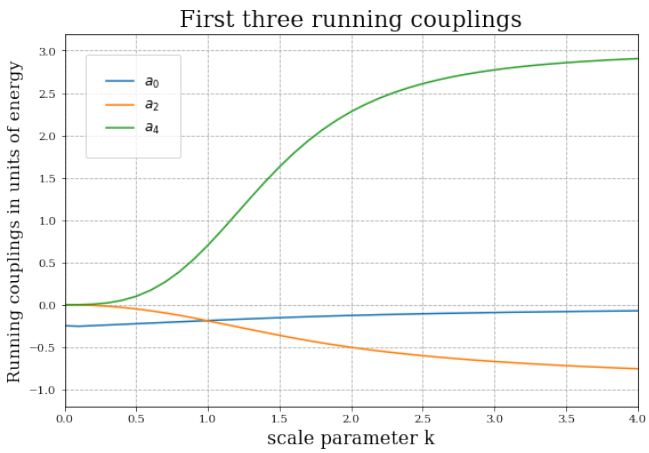


Figure 8.1.: Flow equations for the first three running couplings w.r.t. the energy scale parameter. The effective potential is obtained for $k = 0$.

[1] The axes are sometimes not fully displayed.
[2] We tried to vary $a_4$ below $a_4 = 3$, but the approximation is not powerful enough to resolve this part.

We start with the k-dependent equations for the double well potential. The evolution of the energy scale dependent couplings can be seen in Fig. 8.1. The effective potential is obtained on the very right side of the graph at $k = 0$. Getting the exact values for the final convex potential using Python, the effective and the classical[3] potential can be plotted against the field $\phi$.

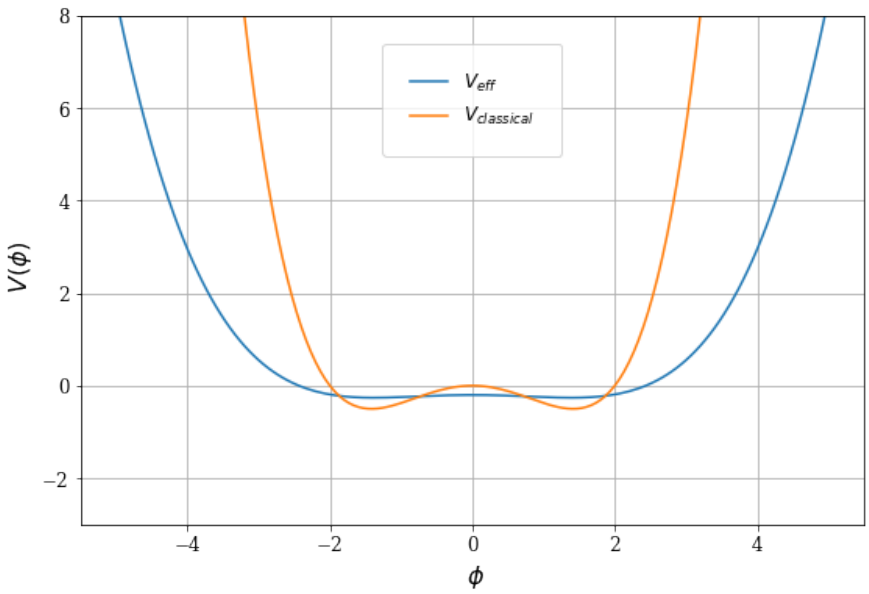


Figure 8.2.: Effective potential for the effective coupling obtained from Fig. 8.1. The symmetry restoring nature of the flow can be seen.

In Fig. 8.2 one can get a good qualitative approximation on what the tendency of the system is, but as we will see in section 8.2, the actual final effective potential looks slightly different. For the one dimensional case, however, we get our final minimum at zero and therefore, this approximation still works well. For dimensions $d \geq 3$ the minima are nontrivial for the final effective potential. Consequently, this simple approximation would not hold if we expand around one minimum but have actually more. What can be done, is to choose a suitable expansion point. That would deliver reliable results if the $\mathbf{Z}_2$-symmetry of the problem is used.
The solution of the solved equations for the RG time can be seen in Fig. 8.3. As expected, the same effective potential as for Fig. 8.1 is obtained. The only difference between the two flow diagrams, Fig. 8.1 and Fig. 8.3, is that for the RG time diagram the evolution for the flow equations has a more natural appearance. For time approaching positive infinity we obtain the final effective potential.

[3]The starting potential, corresponding to the potential of the bare action.

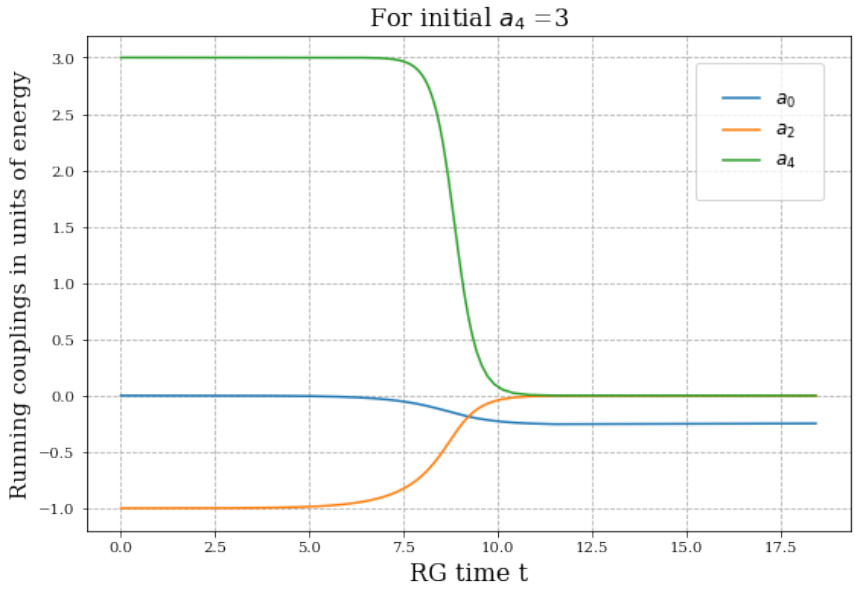


Figure 8.3.: Flow equations for the first three running couplings with respect to the RG time. The effective potential is obtained for $t \to +\infty$.

## 8.2. Solving the PDEs

One obtains a far better solution from solving the simplified nonlinear partial differential equation. Here, the solutions to Eq. 6.5, Eq. 6.7 and Eq. 6.16 are presented.

### 8.2.1. Anharmonic Oscillator

In Fig. 8.4 the solved flow equation for initial value $a_4 = 1$ and negative mass coupling $a_2 = -1$ for the RG time evolution can be seen. We set the UV cutoff, $\Lambda = 1 \times 10^5\,[E]$ and get our effective potential for $t = 14$. It translates to $k = 0.08\,[E]$ for the energy scale. One can run the simulation longer, but the results are not going to change much. The change towards the effective potential happens very fast, starting at approximately $t = 11$, but having in mind the logarithmic relation between $t$ and $k$ it translates to only $\Delta k \approx 1.6\,[E]$. Here, no special care of the boundaries of the differential equation needs to be taken and they can be simply set to zero, as already mentioned in section 7.2, by solving twice as many coupled equations than actually needed.

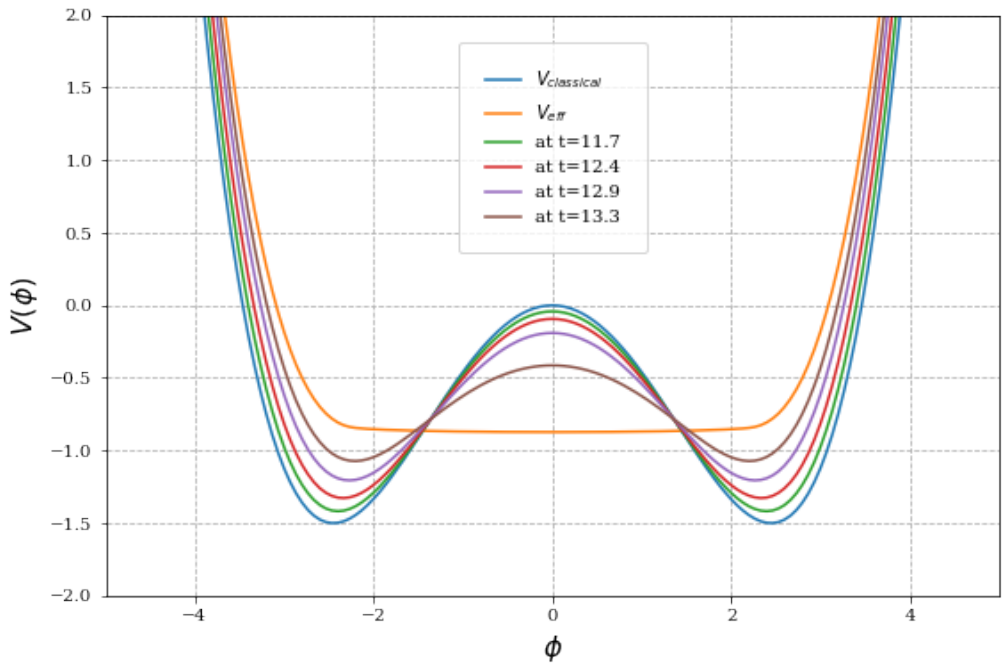


Figure 8.4.: Evolution of the RG time dependent potential of the anharmonic oscillator in $\phi$-convention. One can see for $t = 14$ we get a convex looking potential[4].

We show briefly how the potential looks for $\rho$. Our task is to avoid setting the boundary at $\rho = 0$ explicitly to zero, as we need that value at the end for the minimum

of the effective potential. We use the right-handed derivatives for the first derivative and the central one for the second derivative in Eq. 6.7, so that the flow becomes stable. The potential in Fig. 8.5 is the analog to Fig. 8.4, but for $\rho$-choice of variables. The flow is again convexity restoring and the minimum for the effective potential is at $\rho = 0$. The initial conditions and the parameters are not changed from the $\phi$-computation. The only difference is that $\rho$ goes from 12.5 to 0, because of the used symmetry.

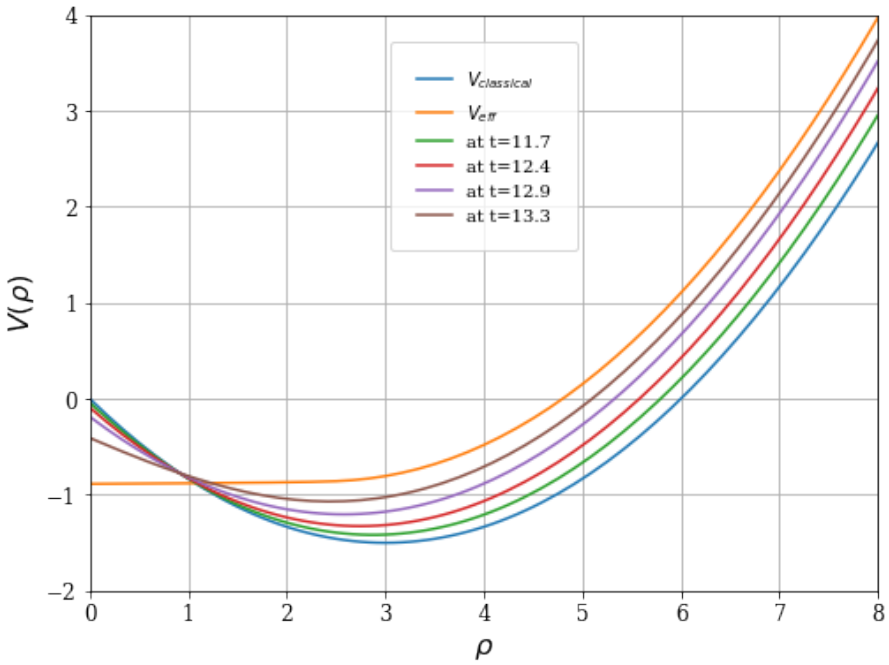


Figure 8.5.: Evolution of the RG time dependent potential of the anharmonic oscillator in $\rho$-convention. One can see that for $t = 14$ we get a flattened potential analog to Fig. 8.4.

The physical interpretation of the effective potential is a special case to the soliton solutions, called instanton. These are tunnelling solutions of the quantum potential which are classically forbidden. The existence of a such solution gives rise to a specific difference between the ground state and the first excited state of the potential. This difference is then computed by searching for the minimum of the effective potential which happens to be our ground state $E_0$[5] so that we can add the square root of the second derivative to this ground state resulting in the first excited state $E_1$. We do not have to set $\phi = 0$ explicitly as we compute the numerical, discrete derivative at the exact point

[5]Note, we did the substraction of the minimum of the potential in section 6.1.

using Eq. 7.8. The final step of the computation is given in the following expression:

$$\Delta E_{gap} = E_1 - E_0 \tag{8.1}$$

### 8.2.2. Energy Gap

As we can see in Fig. 8.6, the obtained results show no significant deviation from the 'exact' solution for large coupling values for $a_4 \equiv \lambda$. However, for smaller couplings, as we see in the logarithmic plot Fig. 8.7, the deviation from the 'exact' values is quite large. This can be explained by our first approximation for the effective action. The shown solution is based on leading order derivative expansion where we do not take into account the possible corrections coming from the wave function renormalisation. Still, the results in the strong coupling region are astonishingly accurate[6].

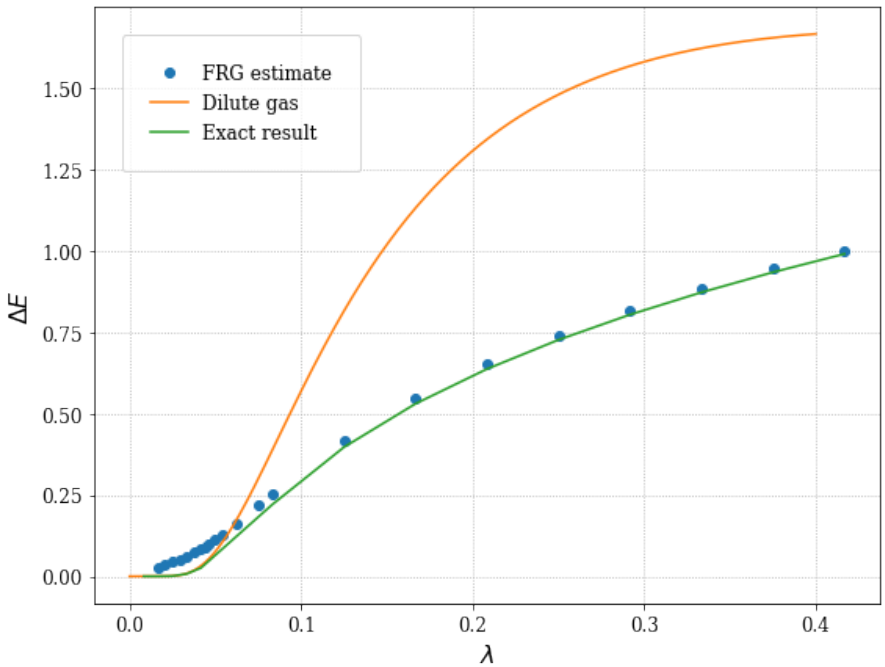


Figure 8.6.: Energy gap for the effective potential of the anharmonic oscillator in $\phi$-convention and RG time. The 'exact' solution is represented by the green graph. The blue points are the FRG estimates and the orange line is the analytic instanton solution. For the larger couplings the FRG estimate shows small deviation from the reference values. The opposite is true for the instanton solution.

For the displayed solution of the analytical computation of the dilute gas assumption we

[6]Note, the coupling $\lambda$ is normalised by a multiplication with 1/24 in Fig. 8.6, Fig. 8.7 and Fig. A.5.

get an accurate description for the weak coupling region. With FRG it is also possible to get good result by going to higher orders in the derivative expansion [Wüs19]. However, for the strongly coupled systems LPA is already sufficient.
The flow is symmetry and convexity restoring. For all of the presented computations there is a clear tendency to be seen. The effective potential represents the new ground state of the system. Areas with larger curvature tend to flow faster, than those with smaller curvature. For $V^{(2)} \to -k^2$ the flow tends to push stronger away from this singularity and hence the symmetry restoring property of the system with $k_2 + V^{(2)} > 0$ is encoded here.

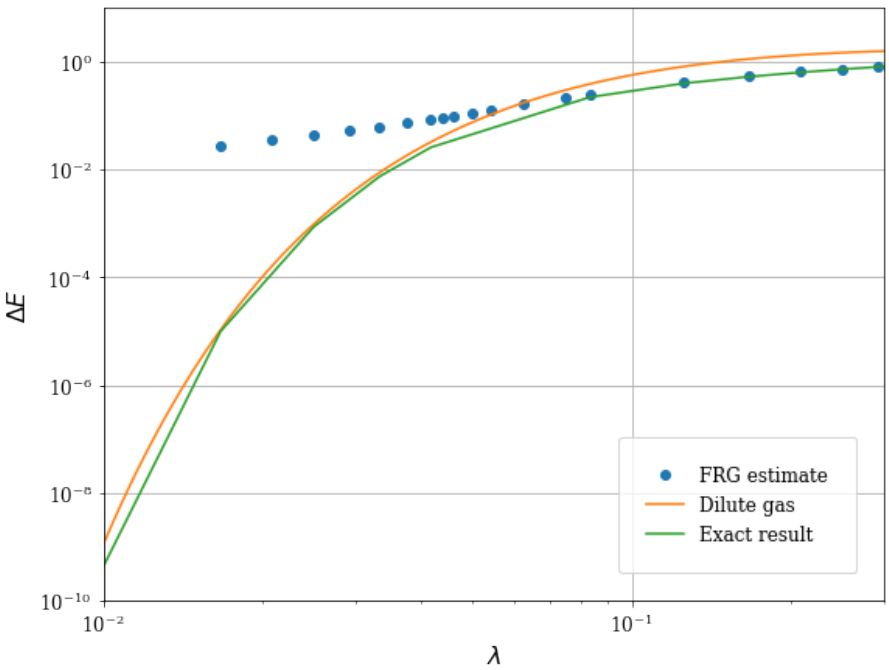


Figure 8.7.: Energy gap in logarithmic plot. The deviation from the 'exact' solution for the weak couplings is more obvious than in a normal scaled plot.

### 8.2.3. Large N Limit $O(N)$ Model

The solution of the one-dimensional flow equation for large $N$ limit $O(N)$ model, given in Eq. 6.16, we obtain by approximating the first derivative of the scale dependent potential by the numerical right handed derivative using the same argumentation as in subsection 8.2.1. The minimum of the effective potential is as well at $\rho = 0$.[7].

[7]For more details the reader is referred to the Mermin-Wagner theorem.

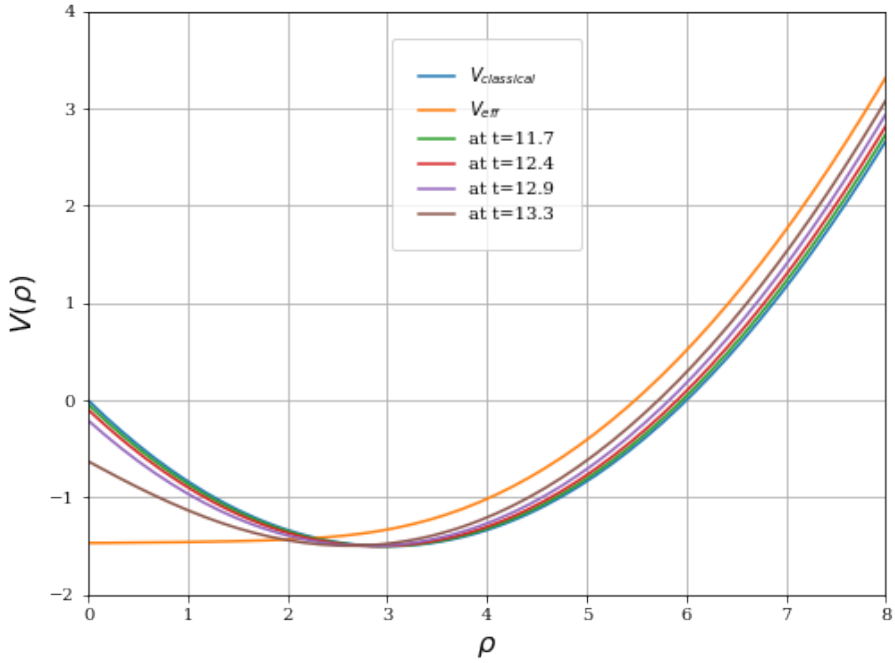


Figure 8.8.: Evolution of the RG time dependent potential of the Large N limit $O(N)$ model in $\rho$-convention. One can see again that for $t = 14$ we get a flattened potential analogously to Fig. 8.4 and Fig. 8.5.

We can look again at the behavior of the system approaching its effective value, as the analytical solution was unfortunately not performed[8]. The result looks similarly to the one obtained for the $O(1)$ model in $\rho$-choice of variables.

[8]There is an analyical solution to the problem using the method of the characteristics for quasi-linear partial differential equations[GW19].

# 9. Conclusion and Outlook

In this thesis we looked at the anharmonic oscillator, as it describes the simplest toy model for symmetry breaking. After that the Large N limit of the $O(N)$ model was investigated. In order to classify the properties of the often used FRG approach for strongly correlated systems, we compared the obtained 'exact' numerical results[1] for the energy gap of the anharmonic oscillator with the FRG estimate.
For the FRG estimate the flow equations are solved using two approaches. The local approximation is done for the $\phi$-dependent effective potential. The results, obtained for this approximation, are good for large coupling values, for smaller - the flow becomes unstable. The other approach is summed up by the finite difference method. Approximating the domain on a lattice and solving the system delivers very good results. The local expansion in comparison to the FDM approach has the advantage of being numerically simpler, but its resolving power is good only for large couplings. With that expansion we checked for some of the couplings, if the energy gap results are correct, and that turned out to be true. Nevertheless, we looked at the energy gap values of the solved PDE, as its resolving power is better.
To leading order derivative expansion of the effective action we get a good approximation for the large coupling values. For the weak coupling region, however, there is a quite large deviation between the 'exact' solution and the FRG estimate solution. We also presented an analytical solution for the small coupling - a dilute gas - assuming non-interacting particles. After comparing the two solutions[2], we see that the deviation for the small couplings is very small, but for large couplings extremely large. This means that for larger couplings the dominance of the interaction in the system increases.
Everything is done in only one dimension and no non-trivial VEV could be obtained. What can be done next, is to look at a case with dimensions more than two. This setup can be used for investigation of phase transitions.
In conclusion, the FRG method is a powerful tool for investigating systems that have no

[1] The solution via diagonalisation of the Schroedinger matrix.
[2] The analytical and the 'exact'.

classical counterpart. These are in general quantum systems that are not in the range of perturbation theory, but by using the non-perturbative FRG approach we can get more insights into how the systems evolve and what the basic properties of going from microscopic theory to a macroscopic one are.

# A. Further Results

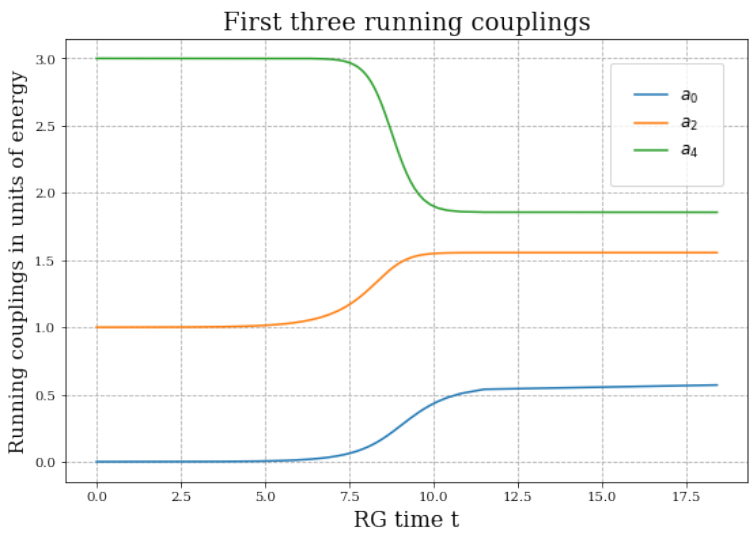


Figure A.1.: RG time dependent flows for the first three equations of the potential, with initial values for the running couplings: $a_0 = 0$, $a_2 = 1$ and $a_4 = 3$.

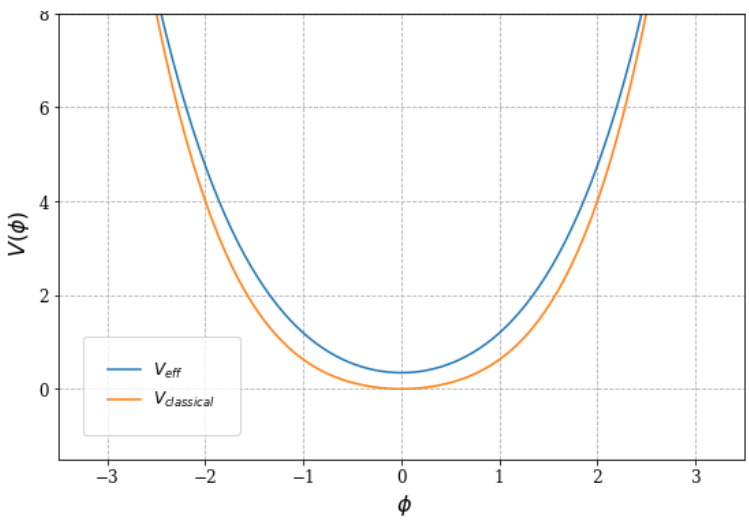


Figure A.2.: Symmetric regime of the potential for initial value of $a_2 = 3$, the corresponding flow is in Fig. A.1.

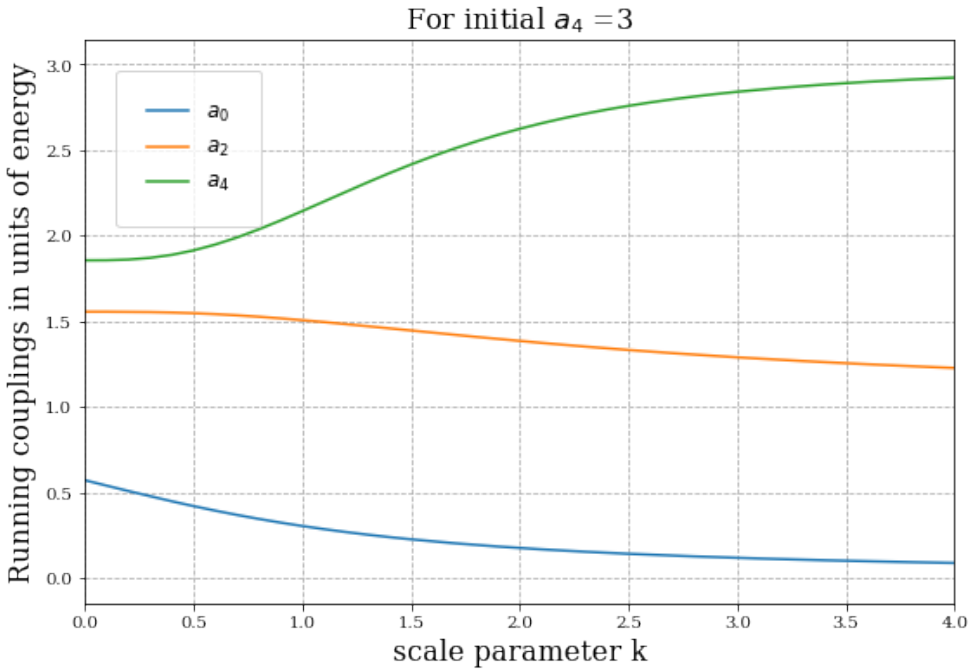


Figure A.3.: Energy scale k dependent flows for the first three equations of the potential, with initial values for the running couplings: $a_0 = 0$, $a_2 = 1$ and $a_4 = 3$.

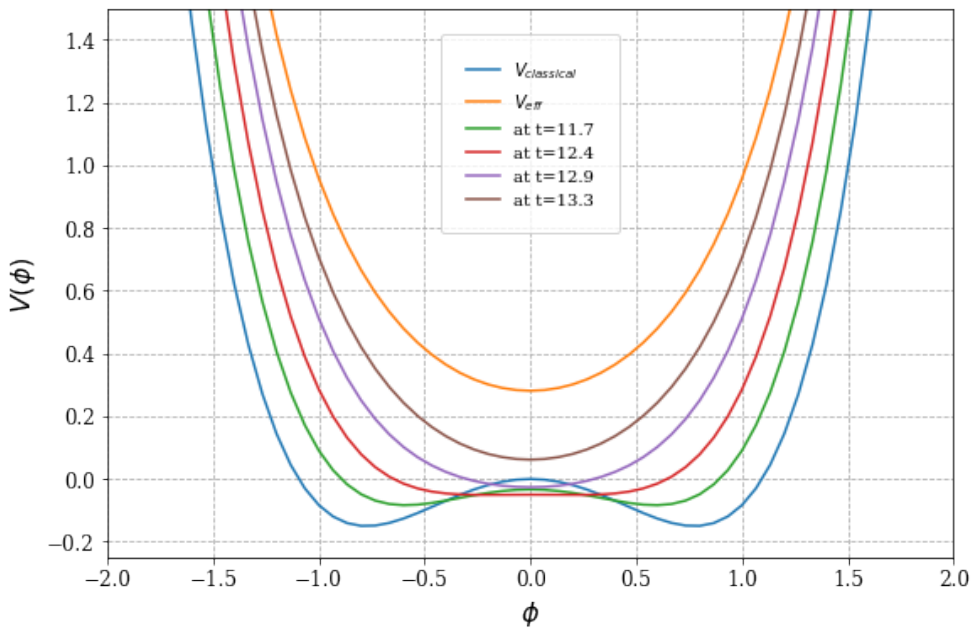


Figure A.4.: Evolution of the RG time dependent potential of the anharmonic oscillator with initial $a_4 = 10$. For $t = 14$ we get the effective potential.

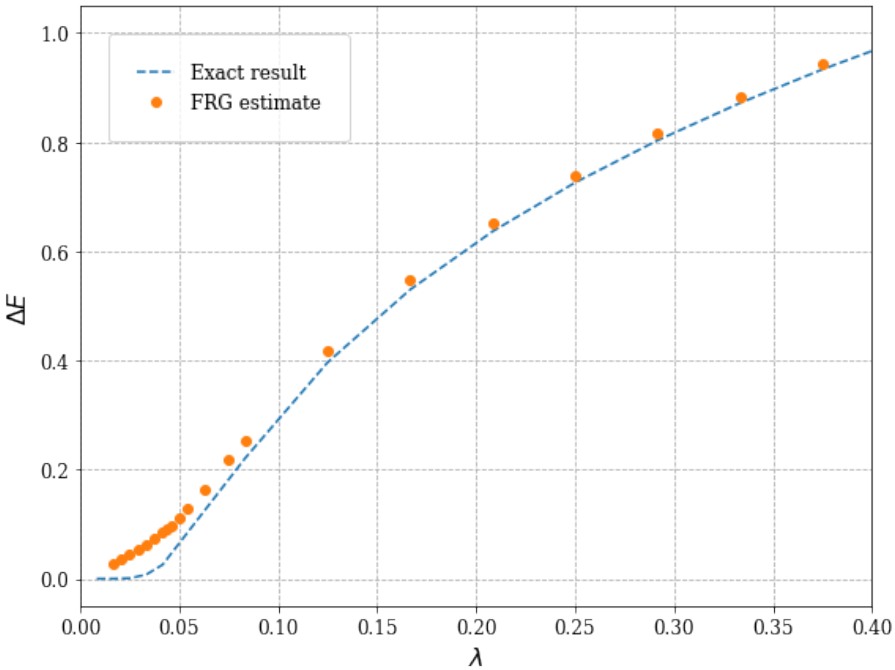


Figure A.5.: Energy gap in normal scaling for the whole range of calculated solutions for the different couplings.

# B. 'Exact' Solution and Dilute Gas

## B.1. Exact Numerical Solution

The numerical solution we obtain by solving the stationary Schroedinger equation and then diagonalising the Hamiltonian matrix. Thus, we are able to find the ground state and the first excited state of the anharmonic oscillator and get the reference values for the energy gap. Here, we show how this is done numerically.
In order to get the expression suitable for numerical implementation we start with the stationary Schrodinger equation

$$E\psi(q) = \hat{H}\psi(q), \tag{B.1}$$

with $E$ being the energy of the corresponding state, $\hat{H}$ the Hamiltonian matrix and $\psi$ the energy state.
The operator $\hat{H}$ in position space is given by

$$\hat{H} = -\frac{1}{2}\partial q^2 + \frac{1}{2}m^2q^2 + \frac{1}{24}\lambda\, q^4. \tag{B.2}$$

We insert the expression for the Hamiltonian into the stationary Schroedinger equation and diagonalise the matrix obtaining an expression suitable for numerical implementation using the discretised derivatives from Eq. 7.8. Thus, we obtain for the second derivative

$$\partial_q^2\psi(q_i) = \frac{\psi(q_{i-1}) - 2\psi(q_i) + \psi(q_{i+1})}{(\Delta q)^2}, \qquad \Delta q = \frac{1}{100},\ q_i \in [-10, 10\,]. \tag{B.3}$$

The stationary equation can be rewritten as

$$\hat{H}\psi(q_i) = \left(\frac{1}{(\Delta q)^2} + \frac{1}{2}m^2q_i^2 + \frac{1}{24}\lambda q_i^4\right)\psi(q_i) - \frac{\psi(q_{i-1})}{2(\Delta q)^2} - \frac{\psi(q_{i+1})}{2(\Delta q)^2}. \tag{B.4}$$

We choose $i \in [0, 2000]$. Now it is easy to rewrite the equation in matrix form and solve it numerically

$$\begin{pmatrix} \frac{1}{(\Delta q)^2} + \frac{1}{2}m^2 q_1^2 + \frac{1}{24}\lambda q_1^4 & -\frac{1}{2(\Delta q)^2} & 0 \\ -\frac{1}{2(\Delta q)^2} & \frac{1}{(\Delta q)^2} + \frac{1}{2}m^2 q_2^2 + \frac{1}{24}\lambda q_2^4 & -\frac{1}{2(\Delta q)^2} \\ 0 & -\frac{1}{2(\Delta q)^2} & \ddots \end{pmatrix} \begin{pmatrix} \psi(q_1) \\ \psi(q_2) \\ \vdots \end{pmatrix} = E \begin{pmatrix} \psi(q_1) \\ \psi(q_2) \\ \vdots \end{pmatrix}. \tag{B.5}$$

For more information look at [Son10]. A complete derivation can be found in [KS20].

## B.2. Dilute Gas Instanton

The analytic dilute gas solution for our purpose can be found in [Zap01]

$$\Delta E = 2\sqrt{2\frac{\sqrt{2}}{\pi\lambda}} exp\left(\frac{-1}{3\sqrt{2}\lambda}\right). \tag{B.6}$$

The dilute gas solution is a good approximation of the weak coupling region of the energy gap curve. The assumption is we have a non-interacting particles, which means we neglect possible occurrence of such processes. We assume that instanton solutions occur not simultaneously and we have no overlap in time for these solutions[Sha14].

# C. Polynomial Expansion in $\phi$ to 12 Order

For RG time: $t = -\frac{k}{\Lambda}$ and $k = \Lambda e^{-t}$.

$$\partial_t a_0 = -\frac{k^3}{\pi}\frac{1}{\left(k^2+a_2-\frac{a_4}{6}+\frac{a_6}{120}-\frac{a_8}{5040}+a_{10}362880\right)}$$

$$\partial_t a_2 = -\frac{k^3}{\pi}\frac{a_4\left(a_4+\frac{a_4}{280}-\frac{a_6}{10}-\frac{a_{10}}{15120}\right)}{\left(a_2+k^2\right)^2\left(a_2+\frac{a_6}{120}+\frac{a_{10}}{362880}-\frac{a_4}{6}-\frac{a_8}{5040}+k^2\right)^2}$$

$$\partial_t a_4 = -\frac{k^3}{\pi}\left(\frac{6a_4^2}{\left(a_2+k^2\right)^3}-\frac{a_6}{\left(a_2+k^2\right)^2}\right)$$

$$\partial_t a_6 = -\frac{k^3}{\pi}\left(-\frac{90a_4a_6}{\left(k^2+a_2\right)^4}+\frac{30a_4a_6}{(k^2+a_2)^3}-\frac{a_8}{(k^2+a_2)^2}\right)$$

$$\partial_t a_8 = -\frac{k^3}{\pi}\left(\frac{2520a_4^4}{\left(a_2+k^2\right)^5}-\frac{1260a_4a_6^2}{\left(a_2+k^2\right)^4}+\frac{56a_4a_8}{\left(a_2+k^2\right)^3}+\frac{70a_6^2}{\left(a_2+k^2\right)^3}-\frac{a_{10}}{\left(a_1+k^2\right)^2}\right)$$

$$\partial_t a_{10} = -\frac{k^3}{\pi}\left(-\frac{113400a_5^5}{\left(a_2+k^2\right)^6}+\frac{75600a_4a_6^3}{\left(a_2+k^2\right)^5}-\frac{3780a_4a_8^2}{\left(a_2+k^2\right)^4}+\frac{90a_4a_{10}}{\left(a_2+k^2\right)^3}-\frac{9450a_4^2a_{10}}{\left(a_2+k^2\right)^4}+\frac{420a_6a_8}{\left(a_2+k^2\right)^3}\right)$$

$$\partial_t a_{12} = -\frac{k^3}{\pi}\left(-\frac{681080400a_4^7}{\left(a_2+k^2\right)^8}+\frac{681080400a_6a_4^5}{\left(a_2+k^2\right)^7}-\frac{37837800a_8a_4^4}{\left(a_2+k^2\right)^6}+\frac{1081080a_{10}a_4^3}{\left(a_2+k^2\right)^5}-\right.$$
$$-\frac{189189000a_6^2a_4^3}{\left(a_2+k^2\right)^6}+\frac{15135120a_6a_8a_4^2}{\left(a_2+k^2\right)^5}+\frac{12612600a_6^3a_4}{\left(a_2+k^2\right)^5}-\frac{252252a_8^2a_4}{\left(a_2+k^2\right)^4}-$$
$$\left.-\frac{270270a_4a_6a_{10}}{\left(a_2+k^2\right)^4}+\frac{6006a_8a_{10}}{\left(a_2+k^2\right)^3}-\frac{630630a_6^2a_8}{\left(a_2+k^2\right)^4}\right)$$

# References


[Aok+98] K. Aokin, A. Horikoshi, M. Taniguchi, and H. Terao. "*Non-Perturbative Renormalization Group and Quantum Tunneling* ". In: (1998). URL: `arXiv:hep-th/9812050v1`.

[GW19] E. Grossi and N. Wink. "*Resolving phase transitions with Discontinuous Galerkin methods*". In: (2019). URL: `arXiv:1903.09503v3%20[hep-th]%203%20Jul%202019`.

[JKM17] D. M. Jackson, A. Kempf, and A.H. Morales. "*A robust generalization of the Legendre transform for QFT*". In: (2017). URL: `arXiv:1612.00462v2`.

[KS20] R. S. Klessen and R. Spurzem. "*Introduction to computational physics*". In: (2020). URL: `https://moodle.uni-heidelberg.de/course/view.php`.

[Lit02] D. Litim. "*Critical exponents from optimised renormalisation group flows*". In: (2002). URL: `arXiv:hep-th/0203006v2`.

[Pav19] P. Pavel. Bachelorthesis: *Investigating the Quantum Mechanical Anharmonic Oscillator with the Functional Renormalization Group*, Supervisor = Prof. Dr. Jan M. Pawlowski. 2019.

[Paw+18] J. M. Pawlowski, J. A. Bonnet, S. Rechenberger, M. Reichert, and N. Wink. "*The Functional Renormalization Group-application to gauge theories and gravity*". In: (2018).

[PJ20] J. M. Pawlowski and J. Jaeckel. "*Advanced Quantum Field Theory lecture*". In: (2020). URL: `private`.

[PS95] M. Peskin and D. Schroeder. *An Introduction to Quantum Field Theory.* United States of America: The Advanced Book Program, Westview Press, 1995.

[Sha14] J. Shapiro. Summary of presentations: *Instantons*, Supervisor = Dr. Philippe de Forcrand. 2014.

[Son10] R. Sonderheimer. Bachelorthesis: *Die Flussgleichung der funktionalen Renormierungsgruppe in der Quantenmechanik*, Supervisor = Prof. Dr. rer nat. Holger Gies. 2010.

[Wip13] A. Wipf. *Statistical Approach to Quantum Field Theory*. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013.

[Wüs19] V. Wüst. Bachelorthesis: *Using the FRG to Solve the Quantum-Mechanical Double-Well Potential Beyound LPA*, Supervisor = Prof. Dr. Jan M. Pawlowski. 2019.

[Zap01] D. Zappalà. "*Improving the renormalization group approach to the quantum-mechanical double well potential*". In: *Physics Letters A* 290.1 (2001), pp. 35–40.

# Acknowledgements and Declaration of authorship

## Acknowledgements

Im Folgenden möchte ich mich bei den Leuten bedanken, die mich auf dem Weg begleitet und unterstützt haben! Ich würde mich zuerst bei Prof. J. M. Pawlowski bedanken, dass ich an diesem spannenden Thema arbeiten durfte. Mein Dank möchte ich auch an meiner Familie und Freunde äußern, die mich unterstützt und das ermöglicht haben. Danke an Nicolas Wink, der trotz der jetztigen Situation, mir hilfreiche Anweisungen während meiner Bachelorarbeit gegeben hat. Vielen Dank speziell an den Leuten, die mir bei dem Korrekturlesen geholfen haben: Nathalie Soybelman, Pavel Popov, Kim Küntzelmann, Jonas Weidner!

## Declaration of authorship

Ich versichere, dass ich diese Arbeit selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe.

Heidelberg, den 03. September 2020,
Anna Radoslavova